UE24MA241B — Linear Algebra and Its Applications

Face Recognition Using Eigenfaces

A Linear Algebra Pipeline: from Pixels to Identity

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Problem Statement & Pipeline Overview

PROBLEM STATEMENT

How can a computer recognize a face it has seen before -> purely using linear algebra?

Given a database of known face images, we want to:

1) **Represent** each image as a vector in a high-dimensional space.
2) **Compress** the space using eigen decomposition.
3) **Project** a new face onto this reduced space.
4) **Identify** it by finding the nearest known face.

LA PIPELINE (9 STEPS)

1. **Matrix Representation**
   Images → flat vectors → data matrix A
2. **Matrix Simplification**
   Mean-centering (baseline removal)
3. **Structure of the Space**
   Rank, column space = 'face space'
4. **Remove Redundancy**
   Linearly independent face directions
5. **Orthogonalization**
   Gram–Schmidt → orthonormal basis
6. **Projection**
   Project face onto face subspace
7. **Least Squares**
   Reconstruct & find best match
8. **Pattern Discovery**
   Eigenvalues of covariance matrix
9. **System Simplification**
   Diagonalization → top-k Eigenfaces

Full Linear Algebra Pipeline

1 Matrix Representation

Images → Row Vectors → Matrix A

2 Matrix Simplification

Mean-center A → $\tilde{A} = A - \mu$

3 Structure of Space

Col space, Rank, Nullity of $\tilde{A}$

4 Remove Redundancy

Linear Independence → Basis

5 Orthogonalization

Gram-Schmidt → Orthonormal Q

6 Projection

Project onto face subspace

7 Least Squares

$\hat{x} = (A^T A)^{-1} A^T b$ → Best match

8 Pattern Discovery

Eigenvalues of $C = \tilde{A}^T \tilde{A}/(n-1)$

9 Diagonalization

$C = PDP^T$ → Top-k Eigenfaces

✓ RECOGNITION OUTPUT

Identified person + reconstruction

Steps 1–3 · Data, Matrix & Space

STEP 1 — Matrix Representation

$A \in \mathbb{R}^{400 \times 4096}$
rows = faces, cols = pixels

Dataset: AT&T/Olivetti Faces — 40 subjects × 10 images = 400 grayscale images (64×64 px).

Each image is flattened: 64×64 = 4096 pixels → 1 row vector of length 4096.

Stack all 400 → Data matrix A of shape 400×4096.

STEP 2 — Matrix Simplification (Mean-Centering)

Compute mean face $\mu = (1/n) \Sigma a_i$ (average of all 400 rows).

Subtract mean from every row: $\tilde{A} = A - \mu$ → this is the 'centered' matrix.

This removes common baseline lighting — only face variations remain.

STEP 3 — Structure of the Space

Column Space:	The 'face space' — directions in which faces actually vary.
Row Space:	Spans the space of meaningful face combinations.
Null Space:	Directions of zero variation — invisible facial features.
Rank–Nullity:	dim(Col $\tilde{A}$) + dim(Null $\tilde{A}$) = 4096 ✓

Steps 4–5 · Basis & Gram–Schmidt Orthogonalization

STEP 4 — Remove Redundancy

Many pixel columns in $\tilde{A}$ are redundant — a dark pixel in one face tends to be dark across all faces.

We use Linear Independence to select only the columns (directions) that are genuinely distinct.

This forms the Basis of the column space of $\tilde{A}$ — a minimal set of vectors that spans 'face space'.

In practice, we extract this basis through the eigen-decomposition in step 8, keeping only top-k directions.

STEP 5 — Gram-Schmidt

Why orthogonalize?

Basis vectors from step 4 may not be perpendicular. Orthogonal vectors make projection simple and noise-free.

Process:

Start with basis vector v_1. Set $q_1 = v_1/\|v_1\|$.

For each v_i, subtract projections onto all previous q_j:

$$v_i' = v_i - \Sigma_j (q_j^T v_i) q_j$$

Normalize: $q_i = v_i'/\|v_i'\|$

Result: Q matrix with $Q^TQ = I$ ✓

Each orthonormal vector = one independent 'facial feature direction' in face space.

Steps 6–7 · Projection & Least Squares Matching

STEP 6 — Orthogonal Projection

Given a new test face f (4096-dim vector),
we project it onto the eigenface subspace Q:

$$f_proj = Q\,(Q^T f)$$

This gives the coordinates of the face in 'eigenface space' — a compact k-dimensional representation instead of 4096.

Faces close in eigenface space belong to the same person.

Projection discards lighting noise and irrelevant pixel variation, keeping only the discriminative structure.

STEP 7 — Least Squares Solution

The system $A \cdot x = f$ (find coefficients to reconstruct f) is overdetermined — more equations than unknowns.

We find the best approximate solution:

$$\hat{x} = (A^T A)^{-1} A^T f$$

The reconstruction of the face is:

$$\hat{f} = A \cdot \hat{x} = \Sigma_i\, x_i \cdot (\text{eigenface } i)$$

Recognition: compute $\hat{x}$ for all training faces and the test face. Find the training face whose $\hat{x}$ is closest (minimum Euclidean distance).

Steps 8–9 · Eigenvalues, Eigenfaces & Diagonalization

STEP 8 — Pattern Discovery via Eigenvalues

Covariance Matrix:

$$C = \tilde{A}^T \tilde{A} / (n-1) \quad \text{shape: } 4096 \times 4096$$

C is symmetric and positive semi-definite.
Eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$ represent variance captured by each eigenvector direction.

Large λ → eigenvector captures strong common facial pattern (e.g. lighting gradient, eye shape).
Small λ → captures noise or minor variation.

Trick: since n=400 << 4096, we compute $C = \tilde{A}\tilde{A}^T$ (400×400) instead — same non-zero eigenvalues, much faster.

STEP 9 — Diagonalization (Eigenfaces)

Decompose C:

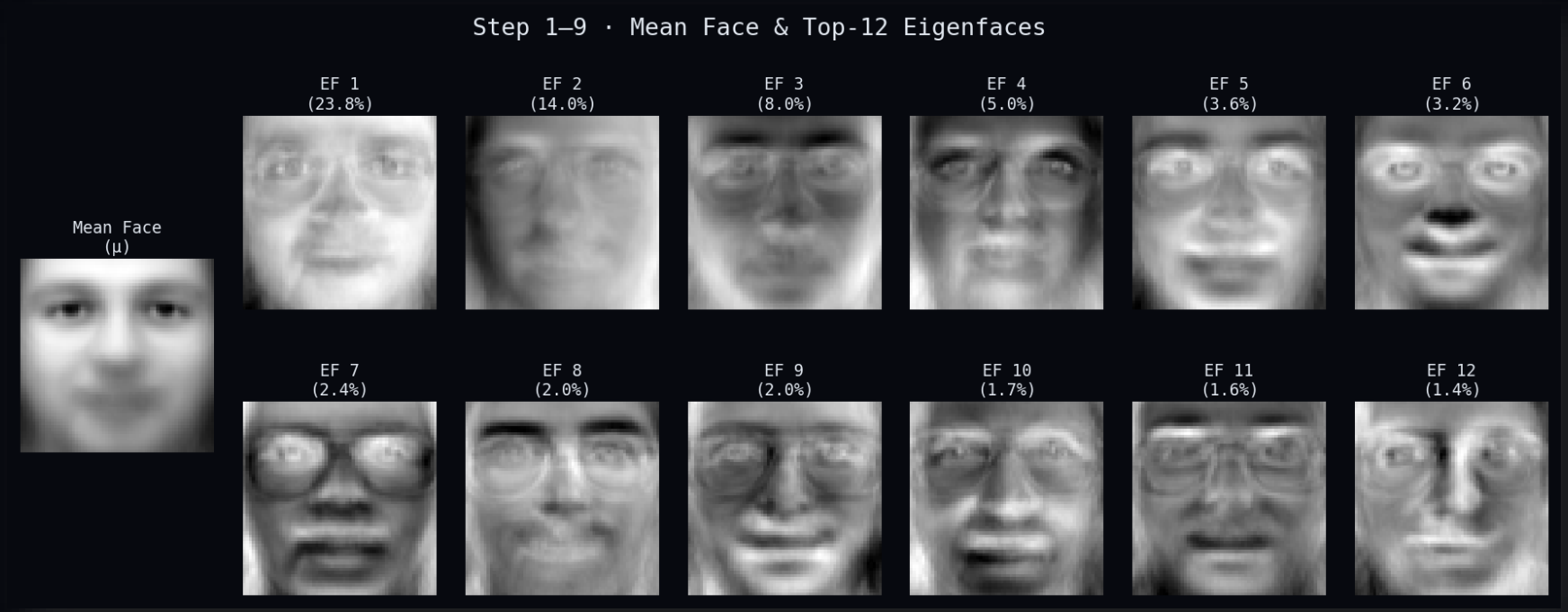
$$C = P\,D\,P^T \quad (C \text{ symmetric} \rightarrow P^T = P^{-1})$$

- P = matrix of eigenvectors (columns = eigenfaces)
- D = diagonal matrix of eigenvalues
- Keep only top-k eigenfaces where cumulative variance ≥ 95%
- Each eigenface is a 'ghost face' — looks like a blurry face encoding one key variation
- Project every face: coords = $P_k^T \cdot \tilde{f}$ (k-dim vector)
- Recognition = nearest neighbor in this k-dim space

Diagonalization compresses 4096D → kD with ≥ 95% variance retained

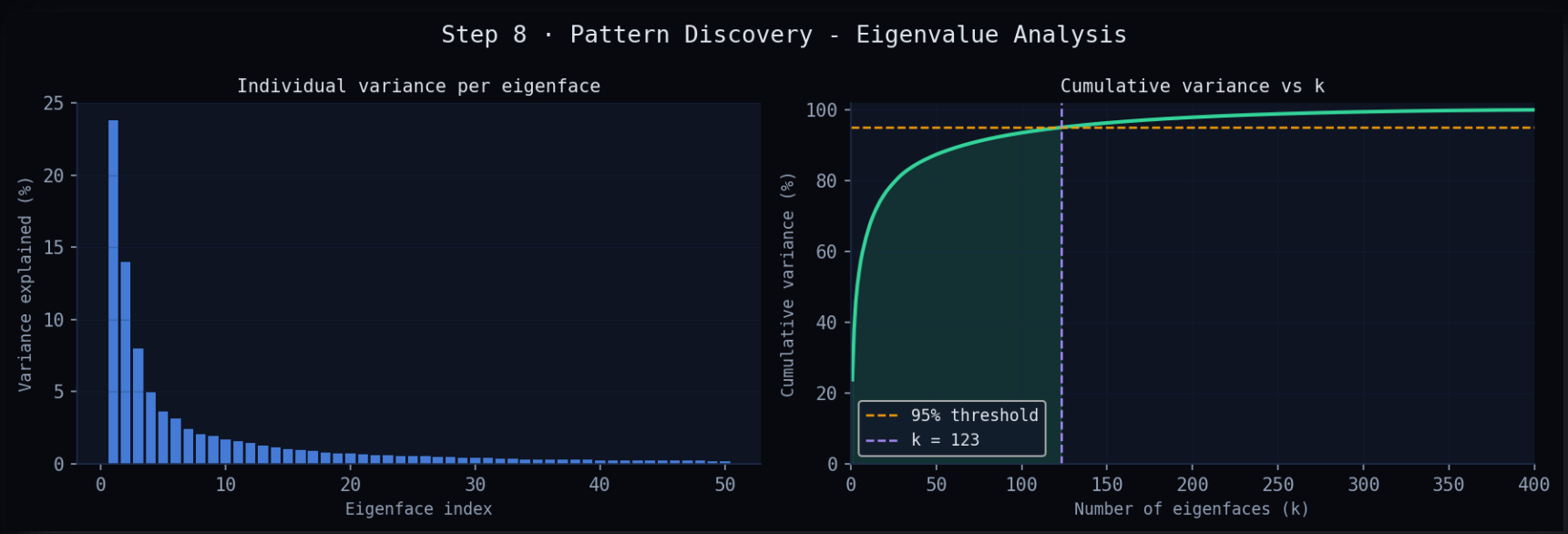
Results

fig1_eigenfaces.png



Results

fig2_variance.png

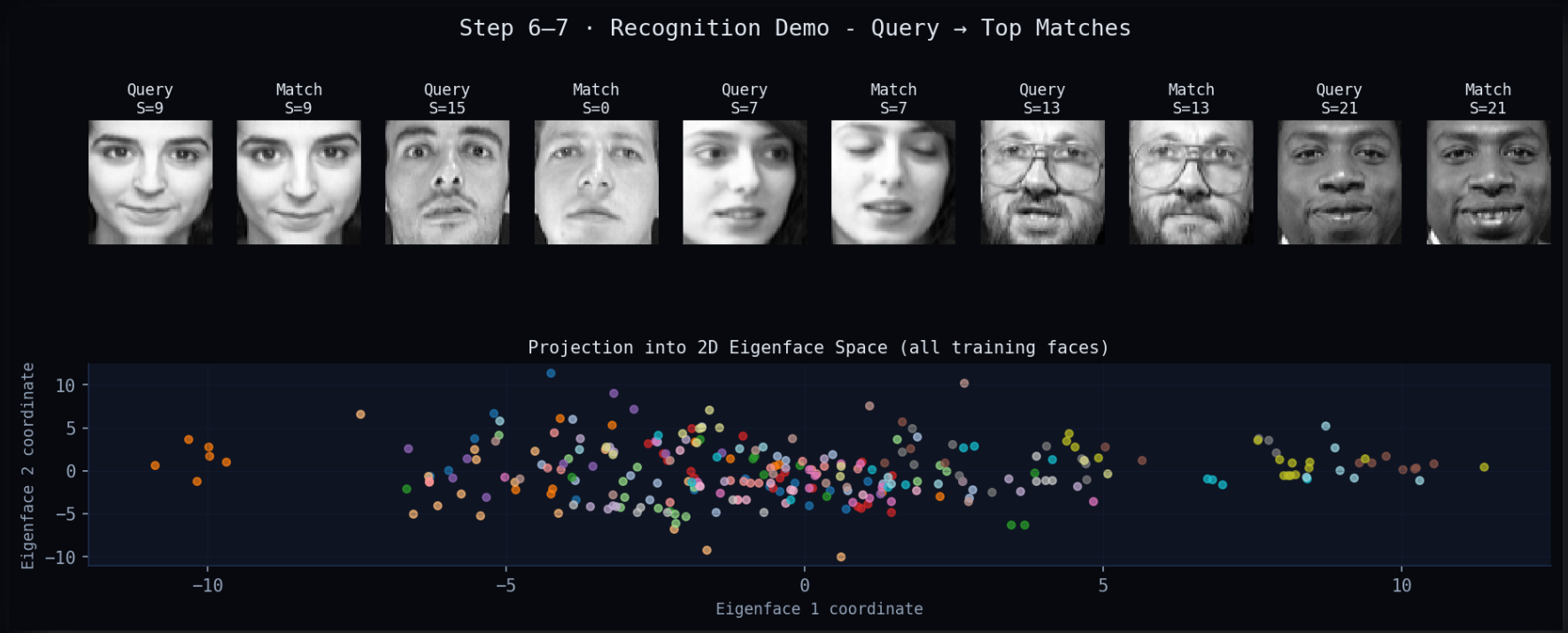


Results

fig3_reconstruction.png

Results

fig4_recognition.png



Results

FINAL OUTPUTS

Mean Face:	Visual average of all training images
Eigenfaces (top-k):	Ghost-like faces = dominant patterns
Variance Plot:	How much info each eigenface captures
Reconstruction:	Original vs eigenface-reconstructed face
Recognition:	Test face matched to correct identity

COMPLETE PIPELINE SUMMARY

Real Data → Matrix A → Center Ã → Basis → Q (G-S) → Project → Least Sq. → Eigenvals → Diag. → Identity